Solution to the motion of the gas

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1 Solution

To find the acceleration of the gas we could firstly find it's speed through Bernoulli's equation:

$$\varepsilon + \frac{P}{\rho} = const \tag{1}$$

where ε is a sum of energy per unit mass of gas.In our problem it consists of two parts:kinetic energy and internal energy,so it means:

$$\varepsilon = \frac{v^2}{2} + \frac{C_V T}{m} \tag{2}$$

From thermodynamics of an ideal gas we know that heat capacity at constant volume equals:

$$C_V = \frac{m}{\mu}\frac{R}{\gamma - 1} \tag{3}$$

Mendeleev Clapeyron's equation for an ideal gas:

$$P = \frac{\rho R T}{\mu} \tag{4}$$

So our first equation becomes:

$$\frac{v^2}{2} + \frac{1}{\mu}\frac{RT}{\gamma - 1} + \frac{RT}{\mu} = const \tag{5}$$

By applying initial conditions we get:

$$v^2 = v_0^2 + \frac{2\gamma R}{\mu(\gamma - 1)}(T_0 - T) \tag{6}$$

Using formula for T from the statement ($T = T_0 + \alpha x$)we get:

$$v^2 = v_0^2 - \frac{2\gamma R \alpha}{\gamma - 1} x \tag{7}$$

We could differentiate our last equation respect to time:

$$2v\frac{dv}{dt} = -\frac{2\gamma R\alpha}{\gamma - 1}\frac{dx}{dt} = -\frac{2\gamma R\alpha}{\gamma - 1}v \tag{8}$$

By simplifying we get our acceleration($\frac{dv}{dt}$):

$$\boxed{a = -\frac{\gamma R\alpha}{\gamma - 1}} \tag{9}$$

To answer for Question 2 we have to use continuity equation for the flow of the gas:

$$\frac{dm}{dt} = \rho S v = const \tag{10}$$

To find $S(x)$ we know that the diameter of the pipe is decreasing linearly so we could write the following:

$$D = D_1 - \frac{D_1 - D_2}{L}x \tag{11}$$

So for the cross-sectional area we get:

$$S = \frac{\pi D^2}{4} = \frac{\pi}{4}(D_1 - \frac{D_1 - D_2}{L}x)^2 = \frac{\pi D_1^2}{4}(1 - \frac{1 - \frac{D_2}{D_1}}{L}x)^2 \tag{12}$$

By using initial conditions and plugging them into the eq(10) we get:

$$\rho(x) = \frac{\rho_0 v_0}{(1 - \frac{1 - \frac{D_2}{D_1}}{L}x)^2\sqrt{v_0^2 - \frac{2\gamma R\alpha}{\gamma - 1}x}} \tag{13}$$

Finally plugging last eq into the eq(4) and using $T(x)$ we get final answer for P:

$$\boxed{P(x) = P_0\frac{v_0}{\sqrt{v_0^2 - \frac{2\gamma R\alpha}{\gamma - 1}}}(1 + \frac{\alpha x}{T_0})} \tag{14}$$